

The method of single points wave spectrum generation for regional sea based on multimodal model in the Bohai sea, China

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ARTICLE INFO

Keywords:

Wave spectrum
Deep learning
Numerical simulation
Multimodal

ABSTRACT

Wave spectra provide comprehensive information on ocean waves, including frequency and direction. However, due to the influence of topographical obstructions and variations in bathymetry, the application of deep learning in wave spectra prediction in semi-enclosed seas is limited. To address this issue, this study proposes a multimodal wave spectrum generation model for single points in regional seas. The model is developed based on an Encoder-Decoder architecture and further enhanced by incorporating the Coordinate Attention mechanism. The dataset for deep learning was simulated using the WAVEWATCH III numerical wave model and validated against buoy measurements. The input data for the deep learning model consists of multi-scale data, including regional wind fields and bathymetry, wind parameters and bathymetry at the target point. To evaluate the model's performance, we assessed the accuracy of the wave spectrum and wave spectrum integral parameters using the correlation coefficient, mean absolute error and root mean square error. Statistical analysis of wave spectrum and wave spectrum integral parameters illustrates that the generated wave spectra accurately reflect the total energy and energy distribution characteristics of the real wave spectra. The results demonstrate that the deep learning model effectively learns the nonlinear relationship between the input data and the wave spectrum in semi-enclosed seas.

1. Introduction

Waves are a common phenomenon in the ocean and are closely related to marine engineering and offshore activities. Therefore, studying ocean waves holds significant social and scientific value. The wave spectrum serves as a tool for describing ocean waves. Compared to wave statistical parameters, the wave spectrum provides a more comprehensive representation of wave energy distribution across different frequencies and directions. Additionally, the wave spectrum allows for the analysis of spectral characteristics and variation trends of the waves. Calculating source terms such as wind input, bottom friction, wave breaking, and dissipation based on the wave spectrum contributes to a deeper understanding of the wave generation processes.

The application of wave spectra in ocean engineering is quite extensive. For instance, wave spectra are utilized for calculating wave loads, assessing structural fatigue, and determining dynamic responses in marine engineering. Additionally, wave spectra are used to calculate the motion and orientation of vessels instead of relying solely on wave statistical parameters. Currently, the primary methods

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for obtaining the wave spectrum are in-situ measurements, remote sensing, and numerical simulations. In situ measurements of wave characteristics are conducted using buoys, vessels, and seabed sensors, which directly observe wave properties to estimate wave spectra. Remote sensing technologies utilize satellite altimetry, synthetic aperture radar (SAR), and LiDAR/radar systems to acquire extensive wave data from a distance. Numerical simulations are based on numerical wave models (such as WAM and SWAN) that calculate wave spectra using parameters like wind fields and water depth. Numerical wave models are established based on the random phase spectral action density balance equation and utilize numerical computational methods to resolve wave spectra. In-situ measurements and remote sensing technologies can be referred to as observation methods. Compared to observation methods, numerical wave simulations are more cost-effective and yield results that are more readily obtainable.

Numerical wave models have advanced to the third generation, with WAM (WAMDI Group, 1988), SWAN (Simulating Waves Nearshore) (Booij et al., 1999) and WAVEWATCH III (Tolman et al., 2002). being prominently recognized on an international scale. Fan et al. (2012) developed a coupled model combining a high-resolution atmospheric model with the WAVEWATCH III numerical wave model, facilitating the simulation of global wave conditions and an evaluation of significant wave heights. Van Vledder and Akpinar (2015) assessed the capability of the SWAN model under both normal and extreme wave conditions using six different wind field products. Majidi et al. (2023) evaluated the uncertainties associated with wave energy estimation using the SWAN model. They proposed improvements by modifying source term parameters within the model, thereby enhancing the accuracy of the average energy output values.

While numerical wave models provide a physical explanation for wave generation processes, their reliance on numerical computation to resolve complex physical dynamics entails significant demands on computational resources. This is particularly evident in high-precision computational scenarios, where even with parallel computing techniques employed to accelerate processing speeds, the resource intensity remains a critical concern for numerical modeling.

The continual advancements in deep learning methods have emerged as potential solutions to the computational intensity of numerical models. Deep learning techniques, with their ability to learn hidden nonlinear relationships through deep network structures, have been extensively applied in oceanographic studies (Pirhooshyaran and Snyder, 2020; Li et al., 2022; Yu et al., 2024). Fan et al. (2020) employed Long Short-Term Memory (LSTM) neural networks to predict significant wave heights using simulated data from the SWAN model, demonstrating that LSTM outperforms traditional machine learning methods in point-based predictions. However, this approach is limited to point-specific forecasts and does not extend to regional wave predictions. Huang et al. (2022) developed a regional wind-wave forecasting model by integrating Convolutional Neural Networks (CNN) with the SWAN model, significantly enhancing computational efficiency. Bai et al. (2022) constructed a two-dimensional wave forecasting model for the South China Sea using a CNN framework, and optimized model hyperparameters with a random search algorithm, achieving favorable results. Yu et al. (2024) develop a spatiotemporal network known as MA-TrajGRU, which integrates a multivariate attention mechanism based on the data simulated from SWAN model. This model was applied to forecast wave parameters during typhoon events in the South China Sea, achieving impressive results that surpassed the performance of commonly used spatiotemporal network models.

In recent years, several studies have explored the use of deep learning for ocean wave spectrum analysis. Han et al. (2022) approached from the perspective of ship motion response, proposing a deep learning method for inverting two-dimensional wave spectra from ship motion responses, and employed a discriminative network to improve the accuracy of the inverted spectra. Nielsen et al. (2024) proposed three machine learning-based methods for sea state estimation using ship response data via the wave buoy analogy, and introduced an uncertainty measure based on the variation among method outputs to identify reliable estimates without requiring ground truth data. As presented in Nielsen et al. (2023), There are a lot of ships at sea, providing sufficient data to support the application of the method. However, these data may still fall short of meeting higher spatiotemporal resolution requirements. Additionally, under extreme wave conditions, ships might be unable to carry out observations. Song and Jiang (2023) investigated the nonlinear relationship between wind fields and wave spectra, restructuring the wind fields into polar coordinates to develop an empirical model for single-point wave spectra in open sea areas using a CNN model. However, this method is ineffective in nearshore and enclosed sea areas. Gao and Jiang (2024) proposed a deep learning downscaling method for wave spectra, utilizing a U-net architecture, and evaluated the model's performance with buoy data and numerical wave model data. Zhang et al. (2024) introduced a method combining a Vision Transformer (ViT) deep learning model with deconvolution blocks to enhance the high precision and wide wavelength range of SWIM spectra, using buoy wave spectra and full wavenumber wind-wave spectra.

The aforementioned research provides valuable insights into the application of deep learning in the study of wave spectra. For instance, it effectively utilizes the nonlinear relationship between wind fields and wave spectra to estimate the wave spectra in open sea areas, achieving commendable results. Additionally, the innovative approach of using vessel motion responses to estimate wave spectra is noteworthy. One of the limitations of this method is the difficulty in obtaining complete observational data under extreme wave conditions. The variability in topography and obstruction makes it challenging for deep learning models to generate wave spectra in semi-enclosed seas. Therefore, it is essential to conduct research on wave spectra specifically for semi-enclosed seas by deep learning, expanding the application of deep learning in the field of ocean waves. This study develops a multimodal wave spectrum generation model for single points within a specific region based on the Encoder-Decoder architecture. By integrating various types of information, such as wind fields and bathymetry, the model leverages the powerful ability of deep learning to capture complex nonlinear relationships to estimate the wave spectrum at the target location. The proposed method is applied to the Bohai Sea of China for validation and practical demonstration. The remaining structure of this paper is as follows. Section2 primarily introduces the data used in this study and validates the results of the numerical wave model. Section3 describes the deep learning model employed in this research, including the input data, processing of the input data, and the specific architecture of the model. Section4 evaluates the performance of the deep learning model and includes a discussion of the results. Section5 presents the conclusion of the paper.

2. Wave simulation

2.1. Study area and data

The aim of this study is to develop a deep learning model for generating regional single-point wave spectra. It is essential to select a suitable regional sea area for the research. The Bohai Sea, a semi-enclosed sea in China, aligns well with the requirements for the regional sea area in this study. To obtain accurate wave simulation data, we expanded the simulation range. Ultimately, the simulation range of the wave numerical model is from 31° to 41° N and from 117° to 137° E. The specific range is illustrated in Fig. 1.

The wind field data is obtained from the 10-meter wind field in the ERA5 reanalysis dataset(<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview>, accessed on 15 October 2024). ERA5 is the fifth-generation ECMWF reanalysis for global climate and weather, covering the past eight decades, with data available from 1940 onwards. The resolution of the wind field is $0.25^{\circ} \times 0.25^{\circ}$. The bathymetry is sourced from the ETOPO1 topographic data published by the NGDC(<https://ngdc.noaa.gov/mgg/global/relief/ETOPO1/tiled/>, accessed on 15 September 2023), covering a global range from 180° W to 180° E and from 75° S to 75° N, with a resolution of $0.1^{\circ} \times 0.1^{\circ}$. The bathymetry of the simulation area in this study is depicted in Fig. 1. Points A and B, highlighted in red in Fig. 1, denote the locations of the deployed buoys. Wave effective significant heights were derived from observations at these buoy sites over the period from January 1, 2021, to December 31, 2021. The obtained observational dataset serves as a critical validation benchmark for the wave numerical model, facilitating the assessment of the model's simulation accuracy and reliability.

2.2. Numerical wave model

The foundation of deep learning lies in having a substantial dataset, ensuring that deep learning methods can capture nonlinear relationships within the data. To obtain the wave spectrum data necessary for this study, we opted to simulate the wave conditions in the target area using a numerical wave model to create the required dataset. After careful consideration, we selected the third-generation wave model WAVEWATCH III as the base model for generating the wave dataset for this research. Inspired by the WAM model (Komen et al., 1994), WAVEWATCH III is a third-generation wave model developed by the US National Centers for Environmental Prediction (NOAA/NCEP), building on the foundations of WAVEWATCH I & II (Tolman, 1991, 1992). As one of the leading international wave models, WAVEWATCH III demonstrates excellent performance in both mesoscale and offshore areas. It is widely utilized in meteorological services, ocean monitoring, and marine research (Barbariol et al., 2017; Cicon et al., 2023; Zheng et al., 2016). With continuous updates and iterations, the WAVEWATCH III ecosystem has significantly improved its compatibility with various file formats, which is another key reason for choosing this model.

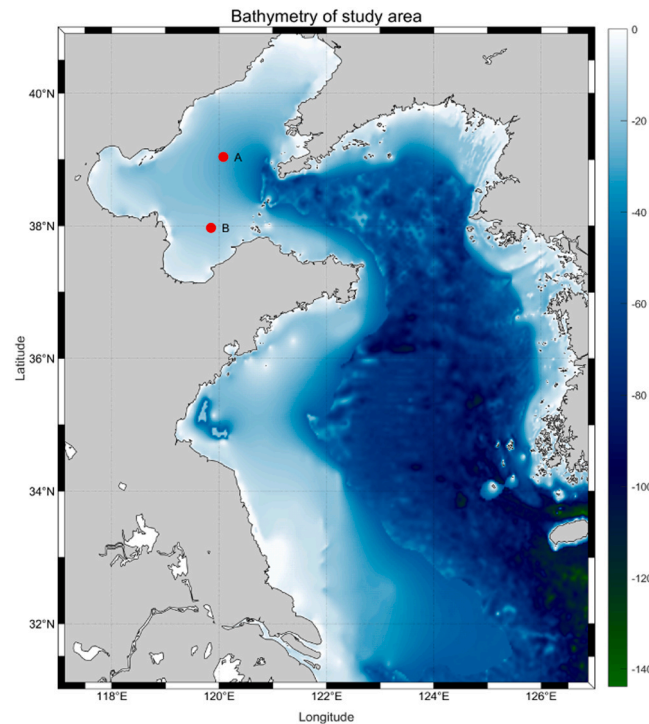


Fig. 1. Bathymetry of the study areas. The red markers indicate the locations of the buoys which are utilized to validate the accuracy of the simulations conducted using the WAVEWATCH III.

The governing equations of WAVEWATCH III in spherical coordinates, defined by longitude (λ) and latitude (ϕ), are as follows:

$$\frac{\partial N}{\partial t} + \frac{1}{\cos \phi} \frac{\partial}{\partial \phi} \dot{\phi} N \cos \theta + \frac{\partial}{\partial \lambda} \dot{\lambda} N + \frac{\partial}{\partial k} \dot{k} N + \frac{\partial}{\partial \theta} \dot{\theta}_g N = \frac{S}{\sigma} \quad (1)$$

$$\dot{\phi} = \frac{c_g \cos \theta + U_\phi}{R} \quad (2)$$

$$\dot{\lambda} = \frac{c_g \sin \theta + U_\lambda}{R \cos \phi} \quad (3)$$

$$\dot{k} = -\frac{\partial \sigma}{\partial d} \frac{\partial d}{\partial s} - \mathbf{k} \cdot \frac{\partial \mathbf{U}}{\partial s} \quad (4)$$

$$\dot{\theta} = -\frac{1}{k} \left[\frac{\partial \sigma}{\partial d} \frac{\partial d}{\partial m} + \mathbf{k} \cdot \frac{\partial \mathbf{U}}{\partial m} \right] \quad (5)$$

$$\dot{\theta}_g = \dot{\theta} - \frac{c_g \tan \phi \cos \theta}{R} \quad (6)$$

where, N represents the spectral action density, t denotes time, ϕ represents latitude, λ represents longitude, θ is the direction, k denotes the wavenumber, σ is the frequency, $\dot{\phi}$ represents the component of wave speed in the latitude direction, $\dot{\lambda}$ represents the component of wave speed in the longitude direction, $\dot{k}$ denotes the component of wave speed in the wavenumber direction, $\dot{\theta}$ represents the component of wave speed in the direction, and $\dot{\theta}_g$ denotes the correction term for wave speed in the direction along great-circle propagation. R represents the Earth's radius, $\mathbf{k}$ is the wavenumber vector, d denotes the average water depth, s is the coordinate aligned with the θ direction, and m is the coordinate perpendicular to s , while U_ϕ and U_λ denote the current velocities in the directions of latitude (ϕ) and longitude (λ), respectively. S represents the source term in the governing equation. In this study, the primary components of S include the atmosphere-wave interaction term S_{in} , the nonlinear wave-wave interaction term S_{nl} , the wave breaking term S_{ds} , and the wave-bottom interactions term S_{bot} . Thus, S can be expressed as:

$$S = S_{in} + S_{nl} + S_{ds} + S_{bot} \quad (5)$$

In this study, the selected simulation domain for the numerical wave model covers a range of 117°E to 127°E and 31°N to 41°N, with a resolution of $0.1^\circ \times 0.1^\circ$. The detailed bathymetry and domain are illustrated in Fig. 1. The boundaries of the simulation domain chosen for this study are significantly larger than the research area, thus eliminating the need for boundary conditions in the numerical simulation. A pre-computation period of one month is used to ensure adequate wind field development. The primary objective of the numerical simulation is to provide data support for the neural network model; therefore, a higher resolution was not selected, as the 0.1° resolution is sufficient to meet this requirement.

The specific configuration details are presented in Table 1.

2.3. Validation of numerical model results

In this study, observational buoy data were utilized to validate the simulation results of the WAVEWATCH III. The locations of the observational buoys are indicated by the blue points A and B in Fig. 1. The observation period for the buoy data was during the year 2022. To evaluate the performance of the simulation, we employed metrics such as the mean absolute error (MAE), correlation coefficient, and root mean square error (RMSE). These metrics were also used to assess the results of the wave spectrum generation model. The formulas for these error metrics are presented as follows.

Table 1
Parameterized configuration in WAVEWATCH III.

Attribute	Model parameters
Grid setup	Resolution $0.1^\circ \times 0.1^\circ$
Bathymetry	ETOPO1
Spectral Action Density Discretization	Number of directions: 24(each with an interval of 15°) Number of frequencies: 30(frequency range from 0.0345 to 0.5473, frequency increment factor $X_\sigma = 1.1$)
Time step	Maximum global time step, maximum CFL time step for x-y and k-theta is 600 s
Initial conditions	Fetch-limited JONSWAP
Nonlinear wave-wave interactions	Discrete interaction approximation (DIA) (Hasselmann et al., 1985; Hasselmann and Hasselmann, 1985)
Wind input	(Rogers et al. 2012; Zieger et al. 2015)
Whitcapping Dissipation	(Rogers et al. 2012; Zieger et al. 2015)
Bottom friction	JONSWAP bottom friction (Tolman, 1991)

$$R = \frac{\sum_{i=1}^n (y_m^i - \bar{y}_m)(y_o^i - \bar{y}_o)}{\sqrt{\sum_{i=1}^n (y_m^i - \bar{y}_m)^2} \sqrt{\sum_{i=1}^n (y_o^i - \bar{y}_o)^2}} \quad (8)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_m^i - y_o^i| \quad (9)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_m^i - y_o^i)^2} \quad (10)$$

where, y_m^i represents the model simulation results, and y_o^i denotes the target values. In numerical simulations, y_o^i refers to the observational buoy data, while y_m^i represents the numerical simulation results. In the deep learning model, y_m^i refers to the results generated by the deep learning, and y_o^i represents the numerical wave model results.

We utilized observational data from the entire year of 2022 to validate the results obtained from the numerical model calculations. Fig. 2 presents the time series comparison of significant wave heights between the wave numerical model and the observational data. As shown in the figure, the simulation results from the numerical model align well with the trends observed in the measured data. This correlation is also evident in the scatter plot in Fig. 3. The calculated correlation coefficients are 0.9 and 0.92, both above 0.9, indicating a strong correlation between the simulation and the observational data. The MAE for the simulated significant wave heights is 0.19 m and 0.19 m, while the RMSE is 0.28 and 0.27. Notably, some zero values appear in the numerical simulation results due to the numerical model assigning a value of zero when the calculated value falls below the minimum threshold. These error metrics indicate that the WAVEWATCH III numerical wave model performs well in simulating this region. Thus, using the wave data set generated by the numerical model as a dataset for deep learning is a reasonable approach.

3. Deep learning method

3.1. Dataset

Considering the influence of various factors on the wave spectrum, we selected target point wind parameters, regional wind field, target point bathymetry, and regional bathymetry as input variables for the model. The selected input data are consistent with those used in the numerical wave model. The model output corresponds to the wave spectrum at the target location.

The dataset for the wave spectrum generation model is derived from the wave spectrum data calculated using the WAVEWATCH III wave model. This dataset encompasses a total of seven years of data, from 2017 to 2023. The total number of samples is 58,788, of which 48,564 (19 points $\times$ 7 years) are training data; 7668 (3 points $\times$ 7 years) are validation data and 2556 (1 point $\times$ 2556) are test data. The locations of the data points are as in Fig. 4. The locations of the data points chosen randomly are shown in Fig. 4, where the red points represent the training data, the blue points indicate the test data and the green points represent the test data. The rationale for randomly selecting data points is that the numerical model is driven by the regional wind field, ensuring consistent physical conditions and constraints throughout the region. This is also the reason why ERA5 wave spectrum data was not used. During the data selection process, the samples were selected using an interval of 24 h. This is to ensure the independence of each wave spectrum.

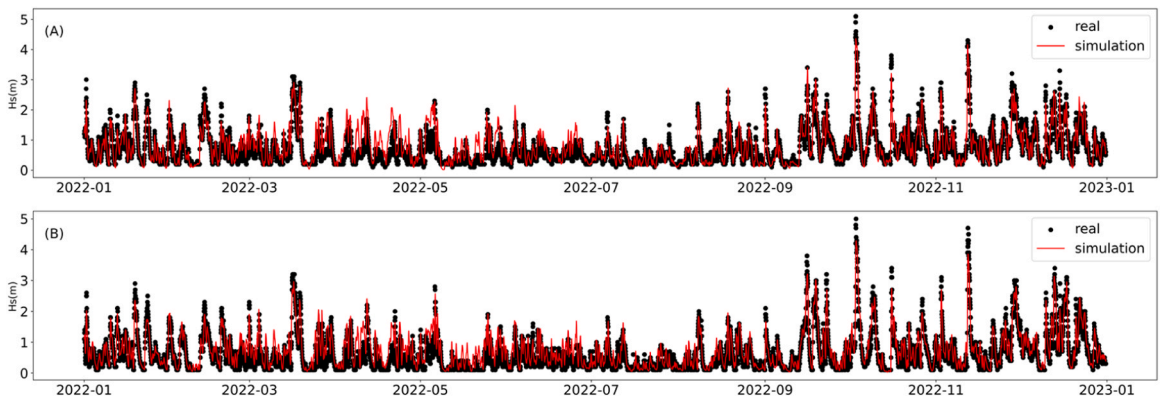


Fig. 2. The time series comparison of significant wave heights (Hs) between the model simulations and buoy observation.

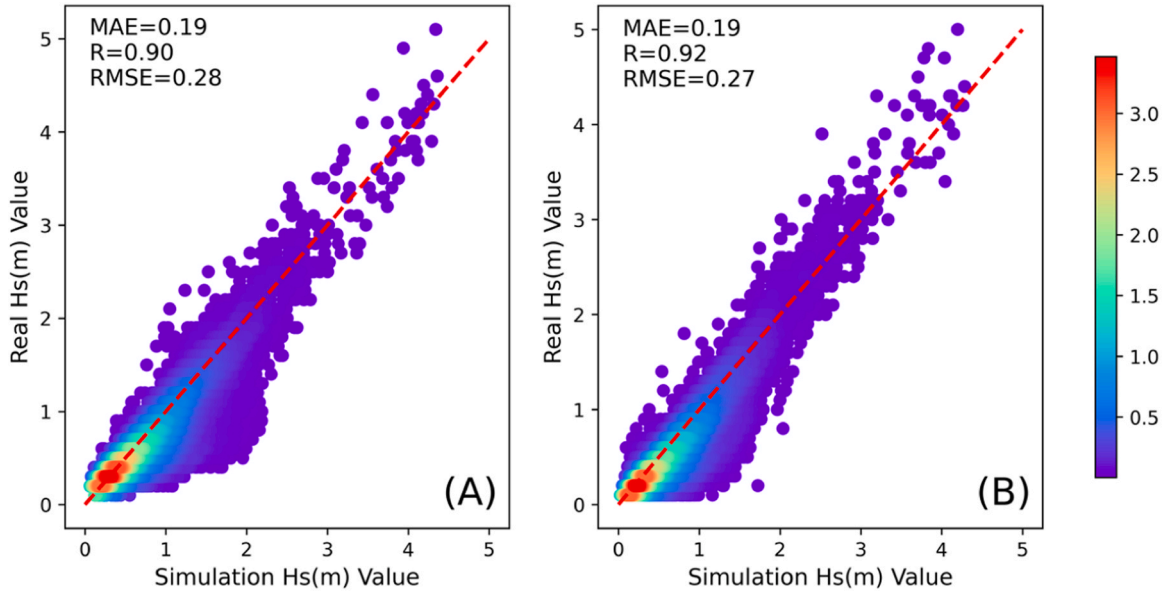


Fig. 3. Scatterplot showing the difference in height of measured and simulated significant wave heights (Hs). The red line indicates a correlation coefficient of 1.

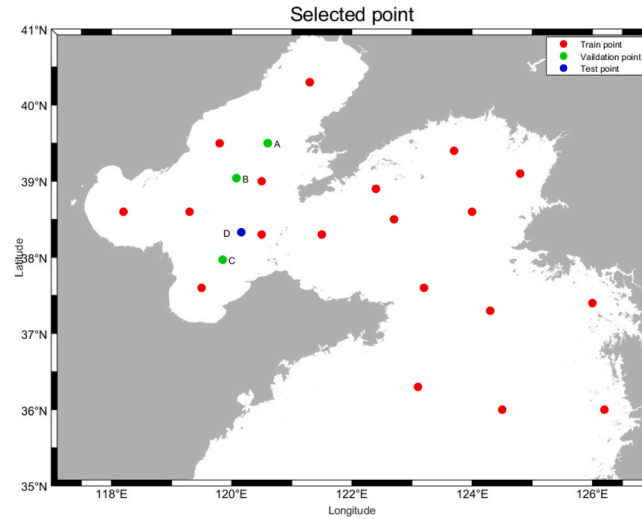


Fig. 4. Selected points for the ocean wave spectrum dataset. The red points represent train samples, the blue points indicate validation samples and the green points represent test sample.

3.2. Data processing

The generation and evolution of ocean waves are influenced by various factors, with wind and bathymetry being the most significant. In numerical modeling of wave processes, wind-driven numerical models are commonly employed. During the simulation, numerical calculation methods utilize iterative computations that incorporate information from different locations and times. Consequently, the wave conditions at a specific location are influenced not only by the local wind but also by the wind conditions across the entire region. The same principle applies to bathymetry. Therefore, when selecting the input for the deep learning model, both the target point wind parameters and the regional wind field are considered. The bathymetry has a significant impact on ocean waves, particularly in areas where deep water transitions to shallow water. Additionally, the obstruction influences the propagation of waves. Therefore, incorporating bathymetry as part of the input data for the model is essential. The input features for the deep learning model include the target point wind parameters, regional wind field, target point bathymetry, and regional bathymetry, while the target output is the wave spectrum at the target location.

Once the input data is determined, the next step is to process the inputs and outputs. It is well-known that deep learning algorithms are highly sensitive to input data, and effective data preprocessing can have a beneficial impact on the results. For the wind field, the regional wind field at a specific moment is determined. For example, Fig. 5(a) represents the wind field at 00:00 on January 1, 2021. If unprocessed wind field data is used to generate the wave spectra at different target points in the figure, the input wind field data will be the same. This situation can affect the model's ability to extract features from the wind field, potentially hindering its effectiveness in capturing the differences and variations among various target points. Therefore, appropriate preprocessing of the wind field to ensure its variability between different target points is a crucial step in enhancing the model's performance. According to Jiang (2022), there is a significant positive correlation between wave energy at a specific location and the projection of the wind field at that point. Therefore, the approach to processing the wind field is to project it onto the target point. The formula for wind field projection is:

$$U_{proj} = U \cos(\theta_U - \theta_A) \delta \quad (9)$$

$$U_{proj}^2 = U^2 \cos(\theta_U - \theta_A) \delta \quad (10)$$

where, U represents wind speed, θ_U denotes the wind direction, θ_A is the azimuth angle from the projection point to the target point. The angles are all in azimuth. δ is the mask, taking values of 0 and 1, where 0 indicates land and 1 represents the ocean. The inclusion of the mask is intended to convert wind speed over non-ocean areas to zero. For example, we want to project the wind from the projection point onto the target point in Fig. 5(b). U denotes the wind speed and direction at the projection point. $\theta = 0^\circ$ corresponds to propagation from south to north and $\theta = 90^\circ$ corresponds to propagation from west to east. By definition, θ_U and θ_A have been labeled in the figure. Substitute wind speed, θ_U and θ_A into Eq. 9, the projection of the projection point to the target point can be calculated. The projected wind field of the region on the target point can be obtained by traversing the points on the whole region.

The processing was performed only on the original wind field from ERA5, without interpolation. We selected the past 12 h as the wind field input. Therefore, within the range of 117°E to 127°E and 31°N to 41°N , the projected wind field data U_{proj} with a resolution of 0.25° has dimensions of $12 \times 40 \times 40$. The U_{proj}^2 has the same dimensions as U_{proj} . By concatenating these two datasets, the final wind field input has dimensions of $24 \times 40 \times 40$.

The energy distribution in the high-frequency range of the wave spectrum is closely related to the wind parameters at that location. Similarly, we process the wind parameters by referring to Eq. 9. For the target location, θ_A in Eq. 9 is zero, which means it becomes the component of wind speed in the horizontal direction. Considering that wind speed also has a vertical component, we include it as an input as well. Therefore, the wind parameters at the target location over the past 12 h, including wind speed, wind speed squared, horizontal wind speed component, and vertical wind speed component, are selected as another input to the model. The single-point wind data is not treated as a one-dimensional vector; instead, it is concatenated along the column direction, resulting in an input dimension of $1 \times 12 \times 4$.

The bathymetry is used as input to account for the influence of bathymetry on the wave spectrum at the target location. The

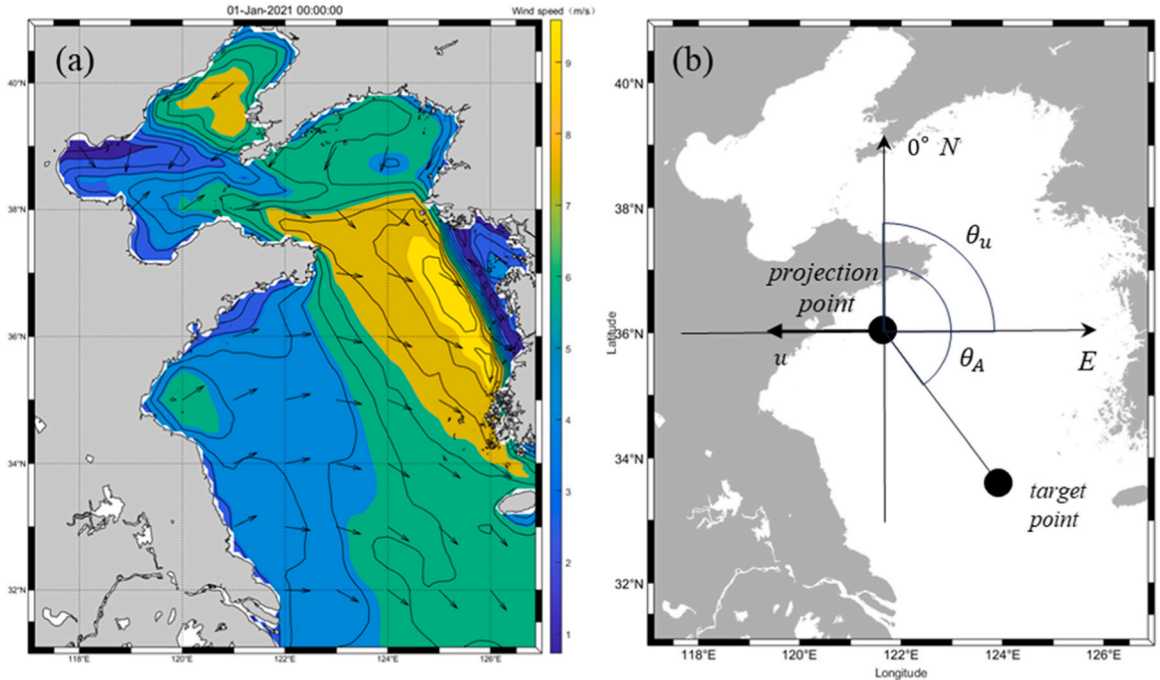


Fig. 5. (a) The wind field at 00:00 on January 1, 2021 in Bohai. (b) A case illustrating how to project wind onto a target point.

bathymetry data is represented in a format of $1 \times 40 \times 40$, while the bathymetry at the target point is expressed as a single value.

3.3. Model structure

In this study, a multimodal wave spectrum generation model was developed. The overall structure of the model is based on the Encoder-Decoder neural network architecture. The model consists of two main components, with the detailed structure illustrated in Fig. 6.

The first part of the model can be considered as feature extraction, where features are extracted from different input data and standardized in size. This part follows a classical Encoder-Decoder architecture. The wind field data, the wind parameters at the target point, and the regional bathymetry each correspond to an Encoder-Decoder neural network structure. These Encoder-Decoder structures are based on Fully Convolutional Networks (FCN). The bathymetry at the target point is handled using a Multilayer Perceptron (MLP). This design is due to the inconsistent dimensions of different input data, making it impossible for a single model to process them. For the wind parameters at the target point, regional wind field, and regional topography, the input format of these data is $C \times H \times W$. The Encoder is adjusted according to the dimensions of each dataset, while a consistent structure is used for the Decoder. The rationale for employing a multimodal approach lies in the significant differences between wind field data and measured water depth data. Even when these datasets are of the same size, simple concatenation can still lead to challenges in feature extraction, not to mention the parameters of target points with inconsistent sizes. The use of a multimodal model offers two main advantages: firstly, it allows for the utilization of data information with different structural forms; secondly, it enables more effective feature extraction from the raw data.

The second part of the model is the feature fusion mapping to generate the wave spectrum. This section employs a U-net model architecture. The U-net neural network, introduced by [Ronneberger et al. \(2015\)](#), enhances the traditional encoding-decoding model by incorporating skip connections. These skip connections cleverly integrate feature information from different levels, thereby improving the accuracy and detail retention of the decoder in image segmentation tasks. It has shown exceptional performance in

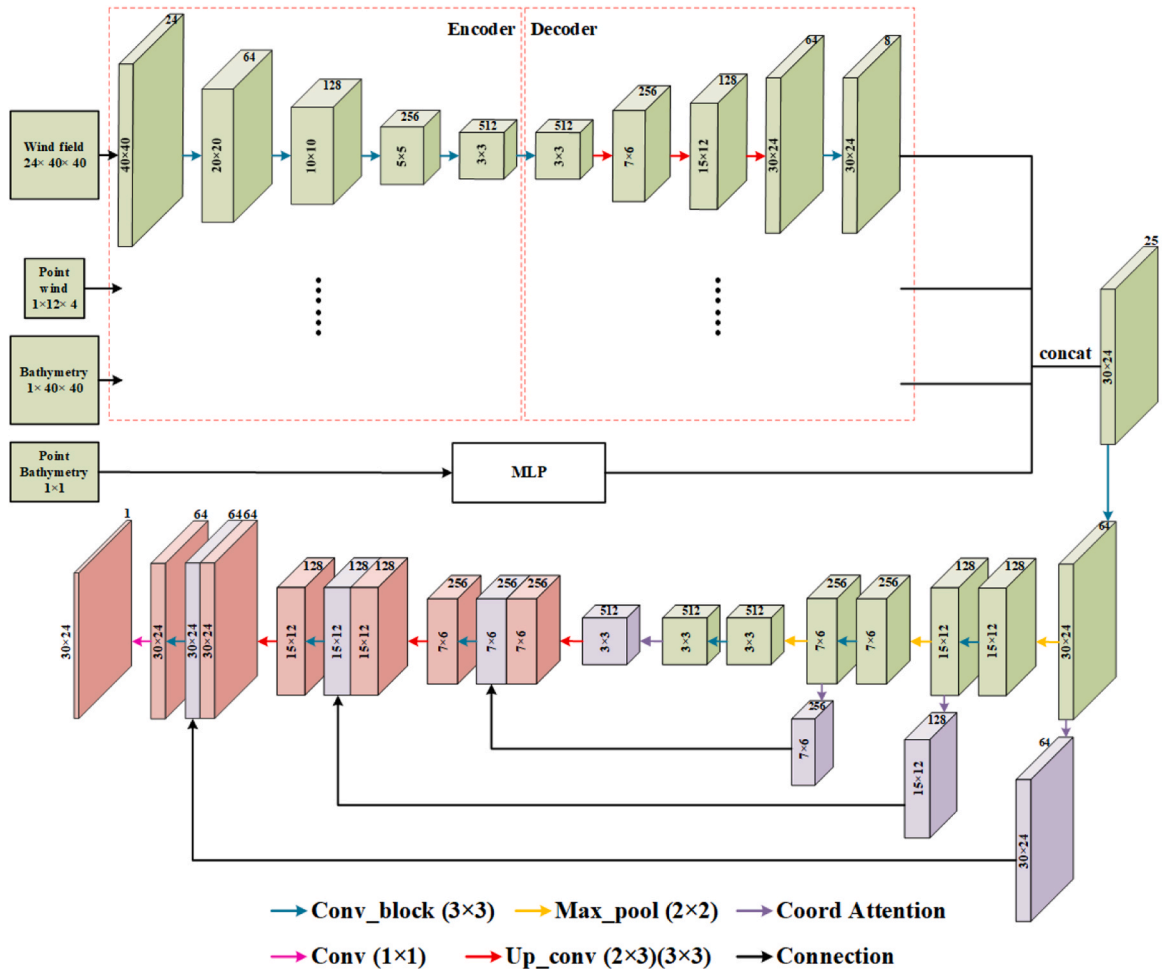


Fig. 6. Schematic diagram of the network architecture.

medical image segmentation applications. Gao and Jiang (2024) utilized the U-net model to establish a wave spectrum downscaling model, demonstrating the feasibility of applying the U-net network to wave spectra. In this research, the U-net neural network is employed as the second part of the model, and it is improved using the Coordinate Attention (CA) mechanism to integrate features from different levels, thus generating the target wave spectrum. The Coordinate Attention mechanism, proposed by Hou et al. (2021), simultaneously takes into account inter-channel relationships and positional information. It not only captures cross-channel information but also incorporates direction-aware and position-sensitive features. This enhancement enables the model to more accurately locate and identify target regions.

The entire model is based on an Encoder-Decoder architecture and can be divided into two sections. The upper part of the figure illustrates a classical Encoder-Decoder model based on a Fully Convolutional Network (FCN), while the lower part shows a U-net architecture, which incorporates skip connections into the Encoder-Decoder framework. Both sections include components such as Conv_block and Up_conv. In the Conv_block, there is a 3×3 convolutional layer, a BatchNorm2d layer, and an activation function. In the first part of the model, the Encoder is adjusted according to the dimensions of the input data, and the Decoder employs a similar structure. A Multilayer Perceptron (MLP) is used for bathymetry at target points, with ReLU selected as the activation function. For the second part, the U-net model utilizes the SiLU activation function. In the downsampling section, the U-net consists of a Conv_block, Max Pooling layers, and a Coordinate Attention (CA) layer. The convolution operations are responsible for extracting data features. Max pooling and increasing the number of channels allow the model to capture higher-dimensional features. The introduction of the CA layer aims to enhance the effective capture of critical features in the feature maps. Unlike other attention mechanisms, the CA layer considers both channel information and direction-related positional information. In the up-sampling section, the number of layers is consistent with the down sampling section. Features extracted by the CA layer are concatenated with the up-sampling section of the model through skip connections. After processing through the up sampling and convolutional layers, a result with dimensions of $64 \times 30 \times 24$ is obtained. Finally, a 1×1 convolutional layer is used to generate the final output. The text within the cuboids represents data dimensions, while the text on top indicates the number of channels.

4. Result and discussion

The version of Python used in this study is 3.9, compiled with the PyCharm IDE, utilizing PyTorch 2.0.1 as the foundational framework for deep learning based on CUDA 11.8 compute platform. The development platform is Windows 11, and the NVIDIA GeForce RTX 4060Ti graphics card is employed for model training. The initial learning rate is set to 0.0001, and an early stopping mechanism is implemented to halt training when the loss no longer decreases.

In the Fig. 7, the blue line represents the training loss, while the red line represents the validation loss. The minimum validation loss occurs at the 88th epoch. As the number of epochs increases, the training loss gradually becomes lower than the validation loss. This is due to the implementation of a learning rate schedule that gradually decreases the learning rate during training, allowing the training loss to continuously decline. To prevent model overfitting, an early stopping strategy was also employed in this study.

5. Result

This study validates the results of the neural network model from two aspects. First, the validation focuses on the wave spectrum.

To compare the spectral characteristics of the wave spectrum, we visualized and compared the wave spectra. The comparison is illustrated in Fig. 8. Fig. 8 presents the comparison of wave spectra across various time in 2022. From the Fig. 8, it can be observed that

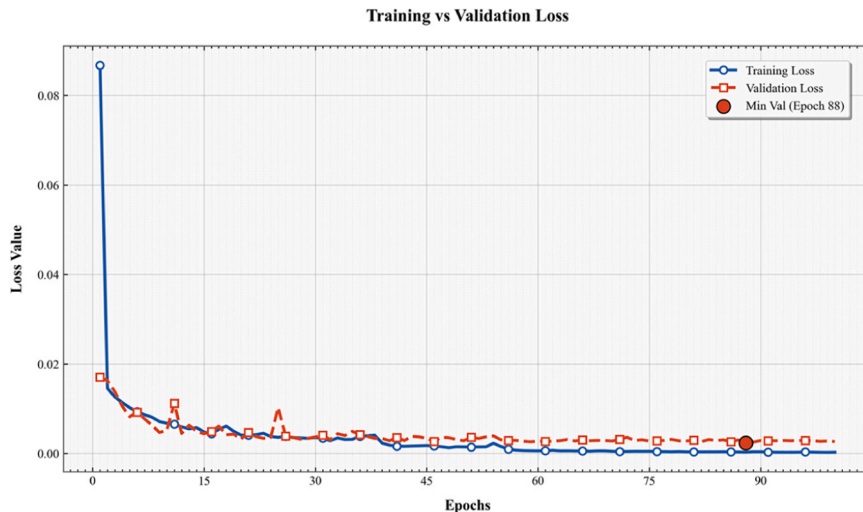


Fig. 7. Loss curve.

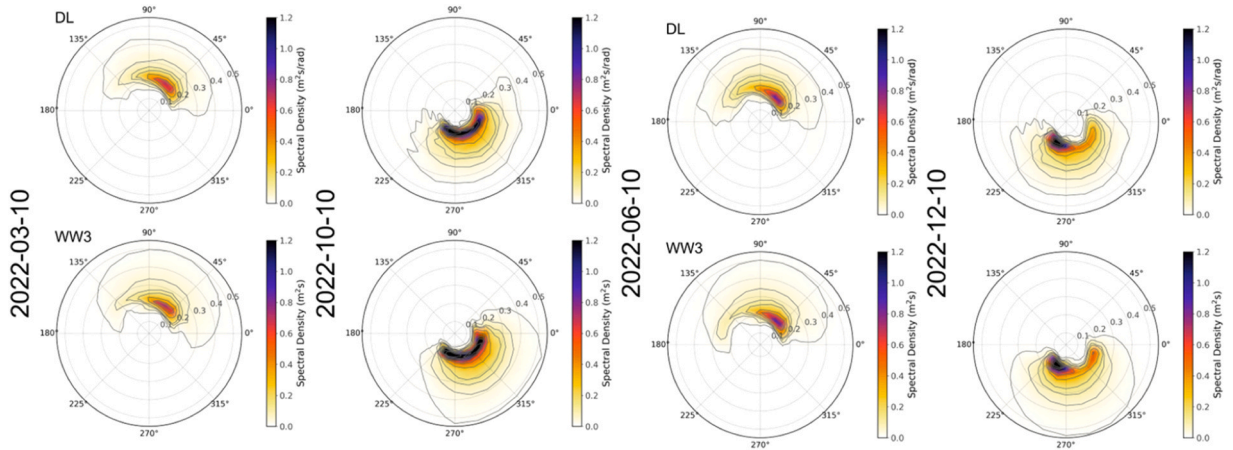


Fig. 8. Comparison between numerical simulations and deep learning wave spectra with different time at Point D in Fig. 4.

most of the wave spectra are single peak. Wind wave spectra are generally single peak, indicating that the simulated region predominantly experiences wind waves. This observation aligns with the characteristic of the Bohai Sea region, where wind waves are predominant. The figure allows for a visual assessment of the differences between the wave spectra generated by the deep learning and the ground truth. In the peak regions of the wave spectra, the deep learning model demonstrates good simulation performance.

To quantify the effectiveness of the deep learning model, we calculated the error between the wave spectrum generated by deep learning and the numerical model wave spectrum. Instead of analyzing the wave spectra statistically according to different frequencies, we converted the 2D wave spectra into scatter points and analyzed them statistically. The error comparison is shown in Fig. 9. The evaluation of each parameter's error was conducted using the Mean Absolute Error (MAE), correlation coefficient (R), and Root Mean Square Error (RMSE). After calculating, the correlation coefficient between the wave spectra generated by deep learning and ground truth is 0.99, the MAE is 0.01, and the RMSE is 0.04. These statistical errors show that there is a good fit between the results of deep learning and the ground truth.

Furthermore, in order to verify the accuracy of the deep learning method in each season, we compared the wave spectra after seasonal averaging, as shown in Fig. 10. From the figure, it can be seen that the wave spectra generated by the deep learning method can better reflect the peaks of the seasonally averaged wave spectra. One of the obvious differences is reflected in the fall. There are three distinct peaks in the real values, while there are only two in the deep learning. Even so, the unlearned peak was the smallest of the three. So, we still think that the results of deep learning are very close to the true values. This suggests that there is a strong nonlinear relationship between the wind in the region and the wave spectra at the target locations, and the deep learning method effectively

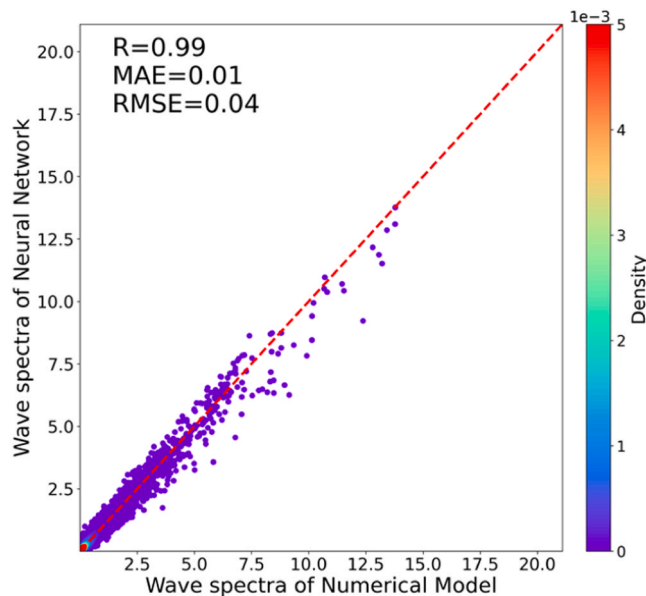


Fig. 9. Scatter plot of wave spectrum between deep learning and numerical model.

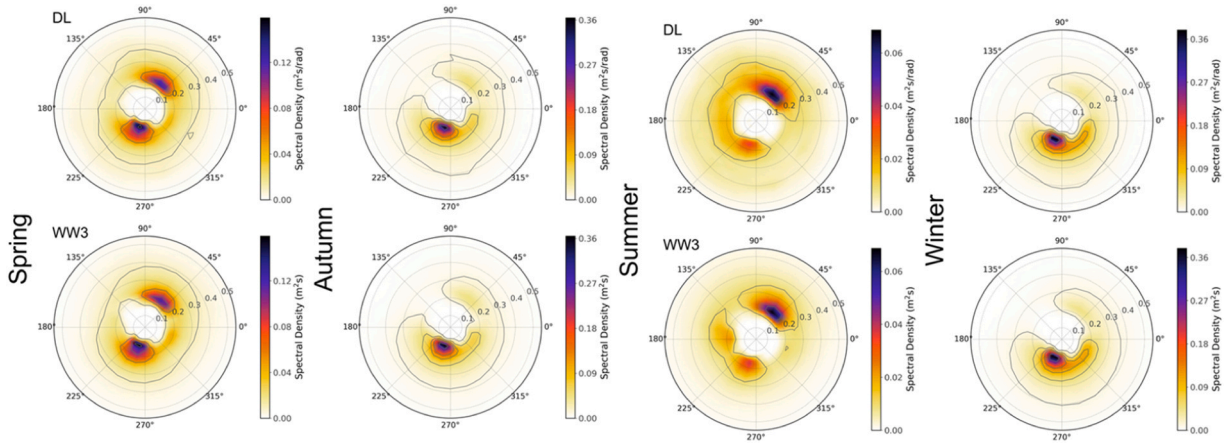


Fig. 10. Seasonal average comparison between numerical simulations and deep learning wave spectra at Point D in Fig. 4.

learns this nonlinear relationship, thereby enabling the rapid generation of wave spectra.

Besides the wave spectrum itself, we are equally concerned about the accuracy of the wave parameters. In this study, we select the three most commonly used wave parameters, namely significant wave height (H_s), mean period (T_m) and mean wave direction (Dir), to further validate the effectiveness of the deep learning. Significant wave height and mean period reflect the overall wave energy, while mean wave direction indicates the energy distribution of the wave spectrum. These parameters allow for a comprehensive assessment of the wave spectrum's energy and distribution. The scatter plots comparing each parameter are shown in Fig. 11, with significant wave height, mean period, and mean wave direction depicted from left to right. The evaluation of each parameter's error was conducted using the correlation coefficient (R), Mean Absolute Error (MAE), and Root Mean Square Error ($RMSE$). In the comparison of significant wave height shown in the first column, the correlation coefficient is 0.99, indicating a high level of consistency between the total energy of the wave spectra generated by the deep learning model and those obtained from numerical calculations. The MAE for significant wave height is 0.07, and the $RMSE$ is 0.09, all of which are relatively low.

The energy consistency of the wave spectra is also demonstrated in the mean period, with correlation coefficients of 0.94. Although slightly lower than those for significant wave height, these values still exceed 0.90, providing further evidence of the high consistency in the total energy of the wave spectra. In the scatter plots for mean wave direction, the correlation coefficients show high consistency, at 0.97. The MAE and $RMSE$ for mean wave direction also remain low, at 12.83 and 28.16 respectively. The statistical errors in mean wave direction indicate that the wave spectra generated by the deep learning model exhibit good consistency in energy distribution, further demonstrating the model's capability to effectively learn the nonlinear relationship between wind and wave spectra.

6. Discussion

In this study, a multimodal wave spectrum generation model based on an Encoder-Decoder neural network architecture was developed and validated in the Bohai Sea of China. The model utilized feature data of various sizes, employing a multimodal processing approach. Through the Encoder-Decoder model, it analyzed and extracted features from multiple data sources, while the U-net model integrated features at different levels, capturing the nonlinear relationship between wind speed, topography, and the wave spectrum. The model was validated using data beyond the training set, and the results demonstrated that the wave spectrum generation model established using deep learning methods can quickly and effectively produce wave spectra for target locations.

Compared to numerical wave models, the deep learning model can rapidly generate wave spectra for target points by learning the nonlinear relationships between parameters and wave spectra, significantly reducing the difficulty of acquiring wave spectra. The validation results in the Bohai Sea region of China were favorable. Given the predominance of wind waves in the Bohai Sea, this model may also be applicable to other wind wave-dominated areas or for numerical wave simulations within certain regions, as these simulations can similarly be wind wave-dominated. This requires further experimental validation.

During the study, we employed a combination of data normalization methods, activation functions, and model structures to develop a sophisticated deep learning model for wave spectrum estimation. It was found that different choices can indeed impact the experimental results, suggesting that uncertainty analysis and optimal parameter selection are worthwhile topics for further research. Additionally, there are areas for improvement in this study. For instance, when using the model to simulate wave spectra at nearshore locations, the error tends to be larger. This may be primarily due to the abrupt changes in nearshore topography altering the dispersion relationship, which deep learning methods cannot effectively learn. Due to significant variations in bathymetry and the complexity of irregular coastlines, many nonlinear coastal wave transformations cannot be adequately resolved by phase-averaged spectral wave models Wang et al. (2022). Introducing phase-resolved models could be an effective approach to improve the accuracy of nearshore wave prediction. Developing separate forecast models for deep and shallow waters might be another solution to this issue. Another potential solution is Physics-Informed Neural Networks (PINN). By solving physical governing equations through derivatives within the deep learning framework, PINN may enhance the accuracy of deep learning in nearshore simulations. This is a direction for future

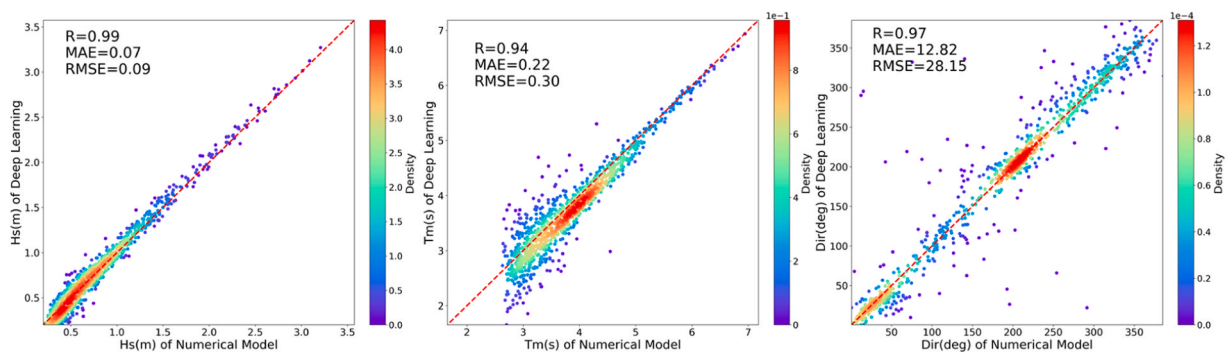


Fig. 11. Scatter plot of wave spectrum integration parameters between deep learning and numerical wave model.

research.

7. Conclusion

This study presents a comprehensive investigation into multimodal wave spectrum generation, with a specific focus on the regional seas, particularly the Bohai Sea in China. By employing the WAVEWATCH III numerical wave model, a robust dataset was constructed, encompassing seven years of wave data from 2017 to 2023. The validation of significant wave height through buoy observations from 2022 demonstrates the reliability of the generated dataset. This work explores wave dynamics in regional seas and aims to expand the application of deep learning in ocean research.

At the core of our research is a multimodal model framework that utilizes an Encoder-Decoder architecture. This framework is divided into two integral components: the feature extraction network, which adeptly manages input data of varying dimensions, and the improved U-Net network. The latter leverages the extracted features to generate accurate wave spectra. Notably, the multimodal processing approach and the enhancement of the U-Net network through the integration of the Coordinate Attention mechanism significantly elevates its performance, allowing for more precise modeling of complex wave patterns. This methodological advancement represents a significant contribution to the field of ocean modeling and deep learning applications.

The effectiveness of the proposed deep learning wave spectrum generation model is quantitatively assessed using three critical evaluation metrics: correlation coefficient (R), mean absolute error (MAE), and root mean square error (RMSE). The comparative analysis of the generated wave spectra against statistical parameters reveals that our model not only successfully captures the essential characteristics of wave spectra in the Bohai Sea but also demonstrates a high degree of consistency with target values. These findings underscore the potential of the developed model to serve as a valuable tool for predicting wave behavior, facilitating better decision-making in marine and coastal management practices.

CRedit authorship contribution statement

Yunqi Lv: Software, Investigation, Data curation. **Xiandong Cao:** Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Data curation. **Zhifeng Wang:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Yuze Niu:** Writing – review & editing, Resources, Investigation. **Ping Wang:** Software, Investigation, Data curation. **Mingli Zuo:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Xianbin Wang:** Writing – review & editing, Resources, Investigation.

Declaration of Competing Interest

We would like to submit the enclosed manuscript entitled ‘The method of single points wave spectrum generation for regional sea based on multimodal model in the Bohai sea, China’, which we wish to be considered for publication in ‘Dynamics of Atmospheres and Oceans’.

We declare that no conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

Acknowledgements

This study was financially supported by the Central Government Funds to Guide Local Science and Technology Development Projects (246Z3501G).

Data availability

Data will be made available on request.

References

- Booij, N.R.R.C., Ris, R.C., Holthuijsen, L.H., 1999. A third-generation wave model for coastal regions: 1. Model description and validation. *J. Geophys. Res. Oceans* 104 (C4), 7649–7666.
- Barbariol, F., Alves, J.H.G., Benetazzo, A., Bergamasco, F., Bertotti, L., Carniel, S., Tolman, H., 2017. Numerical modeling of space-time wave extremes using WAVEWATCH III. *Ocean Dyn.* 67, 535–549.
- Bai, G., Wang, Z., Zhu, X., Feng, Y., 2022. Development of a 2-D deep learning regional wave field forecast model based on convolutional neural network and the application in South China Sea. *Appl. Ocean Res.* 118, 103012.
- Cicon, L., Gemmrich, J., Pouliot, B., Bernier, N., 2023. A probabilistic prediction of rogue waves from a WAVEWATCH III model for the northeast pacific. *Weather Forecast.* 38 (11), 2363–2377.
- Fan, S., Xiao, N., Dong, S., 2020. A novel model to predict significant wave height based on long short-term memory network. *Ocean Eng.* 205, 107298.
- Fan, Y., Lin, S.J., Held, I.M., Yu, Z., Tolman, H.L., 2012. Global ocean surface wave simulation using a coupled atmosphere–wave model. *J. Clim.* 25 (18), 6233–6252.
- Gao, T., Jiang, H., 2024. Statistical Downscaling Of Coastal Directional Wave Spectra Using Deep Learning. *Coastal Engineering*, 104557.
- Group, T.W., 1988. The WAM model—A third generation ocean wave prediction model. *J. Phys. Oceanogr.* 18 (12), 1775–1810.
- Han, P., Li, G., Skjong, S., Zhang, H., 2022. Directional wave spectrum estimation with ship motion responses using adversarial networks. *Mar. Struct.* 83, 103159.
- Hasselmann, S., Hasselmann, K., 1985. Computations and parameterizations of the nonlinear energy transfer in a gravity-wave spectrum. Part I: a new method for efficient computations of the exact nonlinear transfer integral. *J. Phys. Oceanogr.* 15, 1369–1377.
- Hasselmann, S., Hasselmann, K., Allender, J.H., Barnett, T.P., 1985. Computations and parameterizations of the nonlinear energy transfer in a gravity-wave spectrum. Part II: parameterizations of the nonlinear energy transfer for application in wave models. *J. Phys. Oceanogr.* 15 (11), 1378–1391.
- Hou, Q., Zhou, D., Feng, J., 2021. Coordinate attention for efficient mobile network design. *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.* 13713–13722.
- Huang, L., Jing, Y., Chen, H., Zhang, L., Liu, Y., 2022. A regional wind wave prediction surrogate model based on CNN deep learning network. *Appl. Ocean Res.* 126, 103287.
- Jiang, H., 2022. Wind speed and direction estimation from wave spectra using deep learning. *Atmos. Meas. Tech.* 15 (1), 1–9.
- Komen, G.L., Cavaleri, L., Donelan, M., Hasselmann, K., Hasselmann, S., Janssen, P.A.E.M., 1994. Dynamics and modelling of ocean waves. Cambridge University Press, 502p.
- Li, Y., Zhang, X., Cheng, L., Xie, M., Cao, K., 2022. 3D wave simulation based on a deep learning model for spatiotemporal prediction. *Ocean Eng.* 263, 112420.
- Majidi, A.G., Ramos, V., Amarouche, K., Santos, P.R., Das Neves, L., Taveira-Pinto, F., 2023. Assessing the impact of wave model calibration in the uncertainty of wave energy estimation. *Renew. Energy* 212, 415–429, 9.
- Nielsen, U.D., Bingham, H.B., Brodtkorb, A.H., Iseki, T., Jensen, J.J., Mittendorf, M., Takami, T., 2023. Estimating waves via measured ship responses. *Sci. Rep.* 13 (1), 17342.
- Nielsen, U.D., Iwase, K., Mounet, R.E., Storhaug, G., 2024. Uncertainty-associated directional wave spectrum estimation from wave-induced ship responses using machine learning methods. *Ocean Eng.* 313, 119543.
- Pirhooshyaran, M., Snyder, L.V., 2020. Forecasting, hindcasting and feature selection of ocean waves via recurrent and sequence-to-sequence networks. *Ocean Eng.* 207, 107424.
- Rogers, W.E., Babanin, A.V., Wang, D.W., 2012. Observation-consistent input and whitecapping dissipation in a model for wind-generated surface waves: description and simple calculations. *J. Atmos. Ocean. Technol.* 29 (1), 329–346.
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: convolutional networks for biomedical image segmentation. In: *Medical Image Computing and Computer-assisted Intervention—MICCAI 2015: Eighteenth International Conference, October 5–9, 2015, Proceedings, Part III* 18. Springer International Publishing, Munich, Germany, pp. 234–241. *October 5–9, 2015, proceedings, part III* 18.
- Song, Y., Jiang, H., 2023. A deep Learning–Based approach for empirical modeling of Single-Point wave spectra in open oceans. *J. Phys. Oceanogr.* 53 (9), 2089–2103.
- Tolman, H.L., 1991. A third-generation model for wind waves on slowly varying, unsteady, and inhomogeneous depths and currents. *J. Phys. Oceanogr.* 21 (6), 782–797.
- Tolman, H.L., 1992. Effects of numerics on the physics in a third-generation wind-wave model. *J. Phys. Oceanogr.* 22 (10), 1095–1111.
- Tolman, H.L., Balasubramanian, B., Burroughs, L.D., et al., 2002. Development and implementation of wind-generated ocean surface wave models at NCEP[J]. *Weather Forecast.* 17 (2), 311–333.
- Van Vledder, G.P., Akpinar, A., 2015. Wave model predictions in the Black Sea: sensitivity to wind fields. *Appl. Ocean Res.* 53, 161–178.
- Wang, W., Pákozdi, C., Kamath, A., Fouques, S., Bihs, H., 2022. A flexible fully nonlinear potential flow model for wave propagation over the complex topography of the Norwegian coast. *Appl. Ocean Res.* 122, 103103.
- Yu, M., Wang, Z., Song, D., Zhu, Z., Pan, R., 2024. Spatio-temporal ocean wave conditions forecasting using MA-TrajGRU model in the South China Sea. *Ocean Eng.* 291, 116486.
- Yu, M., Wang, Z., Song, D., Cao, X., 2024. Deep learning approach for downscaling the significant wave height based on CBAM_CGAN. *Ocean Eng.* 312, 119169.
- Zieger, S., Babanin, A.V., Rogers, W.E., Young, I.R., 2015. Observation-based source terms in the third-generation wave model WAVEWATCH. *Ocean Model.* 96, 2–25.
- Zhang, R., Qi, J., Yan, Q., Fan, C., Miao, Q., Yang, Y., Zhang, J., 2024. Improvements to the CFOSAT SWIM wave spectrum based on the ViT deep learning model. *IEEE Geosci. Remote Sens. Lett.*
- Zheng, K., Sun, J., Guan, C., Shao, W., 2016. Analysis of the global swell and wind sea energy distribution using WAVEWATCH III. *Adv. Meteorol.* 2016 (1), 8419580.